III Long answer questions

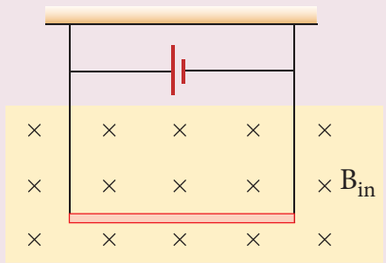
1. Discuss Earth's magnetic field in detail.
2. Deduce the relation for the magnetic induction at a point due to an infinitely long straight conductor carrying current.
3. Obtain a relation for the magnetic induction at a point along the axis of a circular coil carrying current.
4. Compute the torque experienced by a magnetic needle in a uniform magnetic field.
5. Calculate the magnetic induction at a point on the axial line of a bar magnet.
6. Obtain the magnetic induction at a point on the equatorial line of a bar magnet.
7. Find the magnetic induction due to a long straight conductor using Ampere's circuital law.
8. Discuss the working of cyclotron in detail.
9. What is tangent law? Discuss in detail.
10. Explain the principle and working of a moving coil galvanometer.
11. Discuss the conversion of galvanometer into an ammeter and also a voltmeter.
12. Calculate the magnetic field inside and outside of the long solenoid using Ampere's circuital law.

IV. Numerical problems

1. A bar magnet having a magnetic moment $\vec{M}$ is cut into four pieces i.e., first cut in two pieces along the axis of the magnet and each piece is further cut into two pieces. Compute the magnetic moment of each piece.

   Answer $\vec{M}_{new} = \frac{1}{4}\vec{M}$

2. A conductor of linear mass density 0.2 g m^{-1} suspended by two flexible wire as shown in figure. Suppose the tension in the supporting wires is zero when it is kept inside the magnetic field of 1 T whose direction is into the page. Compute the current inside the conductor and also the direction of the current. Assume g = 10 m s^{-2}



   Answer 2 mA

3. A circular coil with cross-sectional area 0.1 cm^2 is kept in a uniform magnetic field of strength 0.2 T. If the current passing in the coil is 3 A and plane of the loop is perpendicular to the direction of magnetic field. Calculate

   (a) total torque on the coil

   (b) total force on the coil

   (c) average force on each electron in the coil due to the magnetic field of the free electron density for the material of the wire is 10^{28} m^{-3}.

UNIT-3(XII-Physics_Vol-1).indd 202 04-03-2019 10:36:11